🟦 Slide 1: Title Slide

Title: Intelligent Driving Decision Strategy for Autonomous Vehicles

Content:

- Project Title: DDS using Machine Learning
- Domain: Artificial Intelligence / Machine Learning
- Developed By: [Your Names]

🟦 Slide 2: Abstract

Content:

- Autonomous vehicles depend on sensor data
- Existing systems consider only external factors
- Proposed DDS considers:
  - External factors (road, pedestrians)
  - Internal factors (RPM, consumables)
- Uses Genetic Algorithm + ML models
- Achieves:
  - 5% lower loss rate
  - Faster decision making
- Cloud-based data processing

👉 Based on abstract in your document

🟦 Slide 3: Introduction

Content:

- Self-driving cars use sensors (GPS, camera, radar)
- Increased sensors → increased data → overload problem
- Overload affects:
  - Speed of decision
  - Vehicle safety
- Solution:
  - Cloud-based DDS system
  - Uses historical data to optimize decisions

👉 From Introduction section

🟦 Slide 4: Motivation

Content:

- In-vehicle computation overload issue
- Need faster decision-making
- Improve driving safety
- Reduce processing burden
- Use cloud computing for efficiency

🟦 Slide 5: Objectives

Content:

- Develop intelligent DDS system

- Analyze internal + external vehicle data
- Improve speed and accuracy
- Use machine learning models
- Optimize driving strategy

🟦 Slide 6: Literature Survey
Content:

- IoT-based self-diagnosis systems
- Deep learning for vehicle condition analysis
- Hidden Markov Models for trajectory prediction
- Neural networks for prediction systems
- Ensemble learning methods

👉 Derived from Literature Survey section

🟦 Slide 7: Existing System
Content:

- Uses ML models:
  - k-NN
  - SVM
  - Random Forest
- Limitations:
  - Lower efficiency
  - Limited accuracy
  - No internal vehicle analysis

🟦 Slide 8: Proposed System
Content:

- DDS with Genetic Algorithm
- Uses:
  - Sensor data
  - Historical cloud data
- Considers:
  - RPM
  - Speed
  - Steering angle
- Benefits:
  - 40% faster than MLP
  - 22% faster than RF
  - Better accuracy

👉 From System Analysis

🟦 Slide 9: System Architecture
Content:

- Steps:

    - Dataset upload
    - Preprocessing
    - Feature extraction
    - Model training
    - Prediction

📌 The diagram on page 16 shows full workflow:

- Input → Preprocessing → ML Models → Prediction → Output

🟦 Slide 10: Data Flow Diagram

Content:

- User uploads dataset
- System processes data
- Algorithms executed
- Results displayed

📌 Refer DFD on page 17 showing:

User → Upload → Process → Execute → Output

🟦 Slide 11: UML Diagrams

Content:

- Use Case Diagram
- Class Diagram
- Sequence Diagram

📌 Pages 19–20 include:

- User interactions
- System structure
- Process flow

🟦 Slide 12: Methodology

Content:

1. Data Collection
2. Data Preprocessing
3. Feature Extraction
4. Train/Test Split
5. Apply Algorithms:
    - Random Forest
    - MLP
    - DDS (Genetic Algorithm)
6. Compare Accuracy
7. Prediction

🟦 Slide 13: Requirements

Content:

Hardware:

- Processor: Intel i3

- RAM: 4GB
- HDD: 250GB

Software:

- Python 3.7
- Anaconda / Jupyter
- Windows/Linux

👉 From Requirements section

🟦 Slide 14: Results

Content:

- DDS outperforms traditional models
- Lower loss rate (~5%)
- Faster decision making
- Efficient driving strategy prediction

🟦 Slide 15: Advantages

Content:

- Faster processing
- Higher accuracy
- Cloud-based computation
- Reduces vehicle overload
- Intelligent decision making

🟦 Slide 16: Conclusion

Content:

- DDS improves autonomous driving decisions
- Combines ML + Genetic Algorithm
- Reduces computation burden
- Enhances safety and efficiency

🟦 Slide 17: Future Scope

Content:

- Deep Learning integration
- Real-time edge computing
- Advanced sensor fusion
- Smart traffic integration
- Fully autonomous optimization

🟦 Slide 18: Q&A

Content:

- Thank You
- Any Questions?